



Republic of Rwanda
Ministry of Education



LSVSS501

SURVEYING OF CIVIL STRUCTURES

Perform Surveying of Civil Structures

Competence

RQF Level: 5

Learning Hours



Credits: 12

Sector: Construction and Building Services

Trade: Land surveying

Module Type: Specific

Curriculum: CBSLSV5002-TVET Certificate V in Land Surveying

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Purpose statement	This module describes knowledge, skills and attitudes required to perform surveying of civil structures. At the end of this module learner will be able to Plan the work, produce plans and maps, perform setting out and to produce as built report.					
Leaning assumed to be in place	Apply land law and regulations, Perform cadastral surveying.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competence and Performance Criteria

Element of Competence	Performance Criteria
1. Plan the work	1.1. Structure elements are well described according to the type of structure.
	1.2. Site reconnaissance is properly conducted according to the procedures.
	1.3. Materials, tools and equipment are correctly selected according to the work to be done.
2. Produce plans and maps	2.1. Surveying data are accurately collected according to the type of structure and project requirements.
	2.2. Surveying data are well processed based on the product requirements.
	2.3. Geometric design is correctly performed according to the type of structure.
	2.4. Plans and maps are properly published according to the product required.
3. Perform setting out	3.1. Road is correctly set out according to its geometric design
	3.2. Dam is correctly set out according to the design
	3.3. Pipeline is correctly set according to the project requirement and site conditions.
4. Perform as built survey	4.1. As built data are properly collected according to the type of structure
	4.2. As built plans and maps are correctly produced according to the type of structure.
	4.3. As built report is properly elaborated based on initial design and executed structure.

Intended Knowledge, Skills and Attitude

Essential Knowledge	Essential skills	Essential attitude
✓ Describing work planning procedures ✓ Identification of surveying software ✓ Identifying surveying tools and equipment ✓ Identifying types of civil structures plans ✓ describing surveying legal aspects ✓ Identifying civil structures setting out process ✓ Describing construction structures ✓ Describing land management principles	✓ Designing work plan ✓ Using surveying Software ✓ Applying surveying computations ✓ Operating surveying equipment ✓ Applying surveying data collections techniques ✓ Analyzing surveying data ✓ Applying cartographic skills ✓ Applying ICT skills ✓ Producing surveying drawings ✓ Interpreting construction drawings and sketches ✓ Applying environmental management skills ✓ Applying reporting skills ✓ Applying communication skills	✓ Having critical thinking ✓ Having Team working spirit ✓ Having responsible ✓ Being Persistent ✓ Having Time management spirit ✓ Being Polite ✓ Being Decisive ✓ Being Research oriented ✓ Being Innovative ✓ Being Flexible ✓ Being Attentive ✓ Being Detail-oriented ✓ Being Prompt ✓ Being Confident ✓ Having Problem solving attitude

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Plan the work 2. Produce Plans and maps 3. Perform setting out 4. Perform as-built survey
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Learning outcome 1: Plan the work

Learning hours: 30 hours

Indicative content

- **Describing Structure elements**
 - ✓ Introduction to the road
 -  Road elements
 -  Sight distance
 -  Road alignment
 - ✓ Introduction to dam
 -  Dam elements
 -  Types of dams

✓ Introduction to pipeline  Pipeline elements  Pipeline alignment • Conducting Site reconnaissance ✓ Prepare for site visit ✓ Conducting site visit ✓ Elaborate site reconnaissance report • Selecting materials, tools and equipment ✓ Tools ✓ Equipment ✓ Materials	
Resources required for the indicative content	
Equipment	GNSS receivers, Camera, Computer, Printer/plotter, PPE
Materials	Pens, note book, flipchart, warning tape, beacon, Building line/ string line, Warning tape, Flags, Paint, Papers.
Tools	Tape measure, Software, Pick axes, Bipod, Umbrella, Bullet, Measuring wheel, Communication toolkit, Ruler, Compass, Ranging rod, A-Frame, Builder's level, Clinometer, Distometer, Area planimeter, Bag and cases, Brush axes, Hand Held, Laser, Hubs, Hummer, machete, Plumb bobs, Saw, Optical squares
Facilitation techniques	Group discussion, scenario, individual activity • Trainer asks the students in small groups to discuss on elements of road, dam and pipeline. • Students present their work and trainer provides feedback and inputs. • Trainer provides a scenario describing the project implementation requirements and its site reconnaissance report, and asks learners to identify activities and allocate resources
Formative assessment methods	• Written assessment • Oral presentation • Performance assessment • Product based assessment

Learning outcome 2: Produce Plans and maps	Learning hours: 30 hours
Indicative content	
• Collecting data of civil structures ✓ Selecting Data Collection Methods  Ground surveying  Aerial surveying ✓ Establishing the Control points	

- ✓ Recording civil structures survey Data

- ✚ Road

- ✚ dam

- ✚ Pipe line

- **Processing data of civil structures**

- ✓ Downloading data

- ✓ Uploading data

- ✓ Cleaning data

- ✓ Analysing data

- **Designing civil structures**

- ✓ Road horizontal alignment

- ✚ Curves

- ✚ Round about

- ✓ Road vertical alignment

- **Publishing Plans and maps**

- ✓ Customise Layout

- ✓ Exporting

- ✓ Plotting

- ✚ Topographic map

- ✚ Layout Plan

- ✚ Longitudinal profile

- ✚ Cross-section profile

- ✚ Earthwork Volume

Resources required for the indicative content

Equipment	Levelling instruments, Theodolites, Total stations, EDM, GNSS receivers, Camera, drones, Laser Scanner, Computer, Printer/plotter, Tacheometer, Drawing table, Wheel barrow, PPE
Materials	Pens, note book, flipchart, warning tape, Nails, Pegs, Building line/ string line, Cement, Gravel, Sand, water, Line chalk, Field Notebook, Warning tape, Flags, Paints, Stake chasers, Papers
Tools	Tape measure, Software, Spades, Hoe, Pick axes, Hammer, Pole, Stands, Bipod, Umbrella, Bullet, Measuring wheel, Communication toolkit, Ruler, Compass, Ranging rod, A-Frame, Builder's level, Clinometer, Distometer, Area planimeter, Bag and cases, Brush axes, Hand Held, Laser, Hubs, Hummer, machete, Plumb bobs, Saw, Optical squares
Facilitation techniques	Group discussion, scenario, individual activity, brainstorming • Trainer asks the learners to brainstorm about the method used in data collection. • Trainer shows the students how to establish control point and allow individual learner to establish its own control point. • Trainer forms a small group and give them a task of collecting data on different structural elements.

	• Trainer shows the students how to process data and asks each learner to process data on computer. • Trainer gives learners within groups a practical exercise about designing horizontal and vertical alignment of road. • Trainer shows students how to publish maps and plans and asks learner to publish plans and map produced during data processing.
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment • Product based assessment

Learning outcome 3: Perform setting out		Learning hours: 30 hours	
Indicative content			
● Setting out road ✓ Selecting road setting out method ✓ Applying road setting out procedures ✚ Horizontal alignment ✚ Vertical alignment ✓ Setting drainage structures ✓ Checking road layer levels ● Setting out dam ✓ Interpreting drawings ✚ Layout Plan ✚ Sections ✓ Setting out dam elements ✓ Checking levels ● Setting out pipeline ✓ Interpreting drawings ✚ Layout plan ✚ Longitudinal profile ✓ Setting out pipeline alignment ✓ Checking levels and slopes			
Resources required for the indicative content			
Equipment	Levelling instruments, Theodolites, Total stations, EDM, GNSS receivers, Tacheometer, Drawing table, Wheel barrow, PPE		
Materials	Pens, note book, flipchart, warning tape, Nails, Pegs, Building line/ string line, Cement, Gravel, Sand, water, Line chalk, Field Notebook, Flags, Paints, Stake chasers, Papers		
Tools	Tape measure, Software, Spades, Hoe, Pick axes, Hammer, Pole, Stands, Bipod, Umbrella, Bullet, Measuring wheel, Communication toolkit, Ruler, Compass, Ranging rod, A-Frame,		

	Builder's level, Clinometer, Distometer, Area planimeter, Bag and cases, Brush axes, Hand Held, Laser, Hubs, Hummer, machete, Plumb bobs, Saw, Optical squares
Facilitation techniques	Group discussion, demonstration, brainstorming • Trainer discusses about the method of setting out and ask the learners to select a method to be used according to given task. • Trainer provides a task to learners within groups of conducting setting out of road and trainer checks the set up work. • Trainer instructs students to perform setting out of different structures based on elements of produced plan or map.
Formative assessment methods	• Written assessment • Oral presentation • Performance assessment

Learning outcome 4: Perform as-built survey	Learning hours: 30 hours
Indicative content	
• Collecting As-built data ✓ Introduction to as-built surveying ✚ Definition ✚ Purpose ✓ Collecting road as-built data ✓ Collecting dam as-built data ✓ Collecting pipeline as-built data • Produce As-built drawings ✓ Road ✚ Layout Plan ✚ Longitudinal profile ✚ Cross-section ✓ Dam ✚ Layout ✚ Profile ✓ Pipeline ✚ Layout Plan ✚ Longitudinal profile • Elaborate as built report ✓ Revising the drawings ✓ Identifying Changes ✓ Providing recommendations	

Resources required for the indicative content

Equipment	Levelling instruments, Theodolites, Total stations, EDM, GNSS receivers, Camera, drones, Laser Scanner, Computer, Printer/plotter, Tacheometer, Drawing table, Wheel barrow, PPE
Materials	Pens, note book, flipchart, warning tape, Nails, Pegs, Building line/ string line, Cement, Gravel, Sand, water, Line chalk, Field Notebook, Warning tape, Flags, Paints, Stake chasers, Papers
Tools	Tape measure, Software, Spades, Hoe, Pick axes, Hammer, Pole, Stands, Bipod, Umbrella, Bullet, Measuring wheel, Communication toolkit, Ruler, Compass, Ranging rod, A-Frame, Builder's level, Clinometer, Distometer, Area planimeter, Bag and cases, Brush axes, Hand Held, Laser, Hubs, Hummer, machete, Plumb bobs, Saw, Optical squares
Facilitation techniques	Brainstorm, demonstration and individual activity • Trainer shows step by step how to collect as-built surveying data, the learner within small group collects that data. • Trainer demonstrates how layout and section of longitudinal profile are produced; trainer instructs students to produce those profile individually. • Trainer allows students to discuss about as-built report and allows each group to present and give them some input about as-built report.
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment • Product based assessment

Integrated/Summative assessment

Integrated situation

Nyanza district wish to construct a road linking road from Kigali to Nyanza with the Kigoma Sector office because the existing one is heavily damaged and causes the problems to the people who needs services from sector's officers. It is in this context, PPD Company Ltd signed a contract with the district to design and set out that road.

As a construction surveying technician from PPD Company Ltd, you are requested to:

1. Produce as-built drawing report for the already laid pipe line along the edge of the road
2. Produce longitudinal profile and cross-section profile of the proposed road, and
3. Set out the centreline of that road.

The road requirements are as follows:

- Length of the road is 100 meter
- Two lanes: 6 meters
- Ditch: 0.5 meters
- Shoulder width: 1.5 meters

Standard of road design and surveying code of practices should be respected. All tools, materials, and equipment are available on-site, and the task should be completed within 10 hours.

Resources

Tools	Flash disk, tape measure, tripod, staff/reflector, club hammer, sledge hammer.
Equipment	Level, total station, DGPS, computer, printer.
Materials/ Consumables	Pipeline original drawings, Pegs, paper, pen, note book, cartridge, Paint, warning tape.

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: Plan the work. (25%)	1.2 Site reconnaissance is properly conducted according to the scope of the work.	1. Site visit is conducted			5
		2. Site boundaries are identified			5
		3. Environmental condition is assessed			3
	1.3 Materials, tools and equipment are correctly selected according to the work.	1. Materials are selected			2
		2. Tools are selected			5
		3. Equipment are selected			5

Produce Plans and maps (30%)	2.1 Data is accurately corrected according to the type of structure	1. Centre line data is recorded			3
		2. Side ditch data is recorded			3
		3. Shoulder data is recorded			3
	2.2 Data is well processed	1. Data is downloaded			3
		2. Software is selected.			3
		3. Data is cleaned			3
		4. Data is analysed			3
	2.3. Plans and maps are properly published according to the product required.	1. Longitudinal profiles are exported			3
		2. Road cross-section profile is exported			3
		3. Layout plan is exported			3
Learning outcome 3: Perform setting out (25%)	3.1 Road is correctly set according to its geometric design.	1. Elements of horizontal curve are well determined			5
		2. Elements of vertical curve are well determined			5
		3. Method of setting out a road is properly selected			5
		4. Procedures of road setting out are accurately applied			5
		5. Road layer levels are well set			5
Learning outcome 4: Perform as-built surveying (20%)	4.1 As-built data is properly Collected	1. Reduced levels are recorded			2
		2. Slope is recorded			2
		3. Dimensions are recorded			2
	4.2 As-built drawings are correctly produced	1. Topographic map is produced			2

		2. Longitudinal profile is produced			2
		3. Layout plan is produced			2
	4.3 As built report is properly elaborated	1. Original drawings are revised			2
		2. Changes/modification are identified			3
		3. Recommendations are provided			3
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References

1. Easa, S. (1988). Selection of Roadway Grades that Minimize Earthwork Cost Using Linear Programming. Transportation Research.
2. Richard Robinson and Bent Thagesen (2004) Road Engineering for Development, second edition: Highway and Traffic Engineering in Developing Countries.
3. AASHTO (1984, 1990) A Policy on Geometric Design of Highways and Streets, Washington, DC: American Association of State Highway and Transportation Officials.
4. Seyed Ataollah et. al. Comparing the expenses of forest road cut and fill operations with standard rules
5. Dr A M Chandra. 2005. Problem solving with theory and objective type question New Delhi' Bangalore ' Chennai